

TSIA2 Math Practice Test 8

PART A — QUESTIONS

TSIA2-M8-001

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A chemistry class needs 2.4 liters of solution for 16 lab stations. At the same rate, how many liters of solution are needed for 30 lab stations?

- A) 3.6 liters
- B) 4.0 liters
- C) 4.5 liters
- D) 5.2 liters

TSIA2-M8-002

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve for x : $9 - 2(x + 4) = 3x - 14$

- A) 1
- B) 3
- C) 5
- D) 7

TSIA2-M8-003

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular storage unit has a floor area of 216 square feet. The length is 18 feet. What is the width of the storage unit?

- A) 10 feet
- B) 12 feet
- C) 14 feet
- D) 16 feet

TSIA2-M8-004

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The number of inches of rain recorded over five months was 2.1, 3.4, 1.8, 4.0, and 2.7. What is the range of the rainfall data?

- A) 1.9 inches
- B) 2.2 inches
- C) 2.7 inches
- D) 4.0 inches

TSIA2-M8-005

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A college bookstore sells a package of 8 pens for \$6.72. What is the price per pen?

- A) \$0.72
- B) \$0.78
- C) \$0.84
- D) \$0.96

TSIA2-M8-006

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A linear function has an initial value of 12 and increases by 5 for each 1-unit increase in x . Which equation represents the function?

- A) $y = 12x + 5$
- B) $y = 5x + 12$
- C) $y = 17x$
- D) $y = 5(x + 12)$

TSIA2-M8-007

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A triangular glass panel has an area of 84 square inches and a base of 14 inches. What is the height of the panel?

- A) 6 inches
- B) 10 inches
- C) 12 inches
- D) 24 inches

TSIA2-M8-008

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A music playlist contains 12 pop songs, 8 rock songs, and 5 jazz songs. If one song is selected at random, what is the probability that the song is rock or jazz?

- A) $\frac{8}{25}$
- B) $\frac{12}{25}$
- C) $\frac{13}{25}$
- D) $\frac{17}{25}$

TSIA2-M8-009

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A water bill decreased from \$96 to \$78. What was the percent decrease?

- A) 12.5%
- B) 15%
- C) 18.75%
- D) 23.1%

TSIA2-M8-010

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A company rents equipment for a flat fee of \$70 plus \$24 per day. If the total rental cost is \$238, how many days was the equipment rented?

- A) 5 days
- B) 6 days
- C) 7 days
- D) 8 days

TSIA2-M8-011

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A circular sign has an area of 64π square inches. What is the diameter of the sign?

- A) 8 inches
- B) 12 inches
- C) 16 inches
- D) 32 inches

TSIA2-M8-012

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The table shows the number of customers served by a coffee cart during four time periods.

8:00–9:00: 26 customers

9:00–10:00: 34 customers

10:00–11:00: 38 customers

11:00–12:00: 42 customers

What is the mean number of customers served per time period?

- A) 32
- B) 34
- C) 35
- D) 36

TSIA2-M8-013

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

An account balance is \$725. A withdrawal of \$180 is made, and then a deposit of \$95 is added. What is the new account balance?

- A) \$450
- B) \$540
- C) \$640
- D) \$1,000

TSIA2-M8-014

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression is equivalent to $(3x + 2)(x - 5)$?

- A) $3x^2 - 13x - 10$
- B) $3x^2 - 15x + 2$
- C) $3x^2 + 17x - 10$
- D) $3x^2 - 10$

TSIA2-M8-015

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

The endpoints of a line segment are $(-6, 4)$ and $(2, -2)$. What is the midpoint of the segment?

- A) $(-4, 1)$
- B) $(-2, 1)$
- C) $(-2, 3)$
- D) $(4, -3)$

TSIA2-M8-016

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A data set contains the values 11, 13, 17, 19, and 25. If the value 40 is added to the data set, which measure will change the most?

- A) Mean
- B) Median
- C) Mode
- D) Minimum

TSIA2-M8-017

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A square courtyard has an area of 110 square meters. Which whole number is closest to the side length of the courtyard?

- A) 9 meters

- B) 10 meters
- C) 11 meters
- D) 12 meters

TSIA2-M8-018

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve the system of equations:

$$x + 2y = 14$$

$$3x - y = 7$$

What is the value of y?

- A) 3
- B) 4
- C) 5
- D) 6

TSIA2-M8-019

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular prism has a volume of 360 cubic centimeters. Its length is 12 centimeters, and its width is 5 centimeters. What is its height?

- A) 4 centimeters
- B) 6 centimeters
- C) 8 centimeters
- D) 10 centimeters

TSIA2-M8-020

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A box contains 3 small batteries, 4 medium batteries, and 5 large batteries. Two batteries are selected at random with replacement. What is the probability that both selected batteries are small?

- A) 1/16
- B) 1/12
- C) 1/9
- D) 1/4

PART B — ANSWER KEY + EXPLANATIONS

TSIA2-M8-001 — Correct: C — Correct Answer: 4.5 liters — Explanation: The rate is $2.4 \div 16 = 0.15$ liter per station. For 30 stations, $30 \times 0.15 = 4.5$ liters.

TSIA2-M8-002 — Correct: B — Correct Answer: 3 — Explanation: Distribute to get $9 - 2x - 8 = 3x - 14$, so $1 - 2x = 3x - 14$. Add 14 and 2x to get $15 = 5x$, so $x = 3$.

TSIA2-M8-003 — Correct: B — Correct Answer: 12 feet — Explanation: For a rectangle, area =

length $\times$ width. Since $216 = 18 \times \text{width}$, the width is $216 \div 18 = 12$ feet.

TSIA2-M8-004 — Correct: B — Correct Answer: 2.2 inches — Explanation: The range is the greatest value minus the least value: $4.0 - 1.8 = 2.2$ inches.

TSIA2-M8-005 — Correct: C — Correct Answer: \$0.84 — Explanation: Divide the total cost by the number of pens: $\$6.72 \div 8 = \0.84 per pen.

TSIA2-M8-006 — Correct: B — Correct Answer: $y = 5x + 12$ — Explanation: The rate of change is 5, and the initial value is 12, so the equation is $y = 5x + 12$.

TSIA2-M8-007 — Correct: C — Correct Answer: 12 inches — Explanation: The area of a triangle is $\frac{1}{2} \cdot \text{base} \cdot \text{height}$. Since $84 = \frac{1}{2} \cdot 14 \cdot h$, $84 = 7h$, so $h = 12$.

TSIA2-M8-008 — Correct: C — Correct Answer: $\frac{13}{25}$ — Explanation: There are $12 + 8 + 5 = 25$ songs total. Rock or jazz songs total $8 + 5 = 13$, so the probability is $\frac{13}{25}$.

TSIA2-M8-009 — Correct: C — Correct Answer: 18.75% — Explanation: The decrease is $\$96 - \$78 = \$18$. The percent decrease is $18/96 = 0.1875$, or 18.75%.

TSIA2-M8-010 — Correct: C — Correct Answer: 7 days — Explanation: Subtract the flat fee: $238 - 70 = 168$. Divide by the daily fee: $168 \div 24 = 7$ days.

TSIA2-M8-011 — Correct: C — Correct Answer: 16 inches — Explanation: The area of a circle is $A = \pi r^2$. If $\pi r^2 = 64\pi$, then $r = 8$. The diameter is $2r = 16$ inches.

TSIA2-M8-012 — Correct: C — Correct Answer: 35 — Explanation: The total is $26 + 34 + 38 + 42 = 140$. Dividing by 4 gives a mean of 35.

TSIA2-M8-013 — Correct: C — Correct Answer: \$640 — Explanation: The new balance is $725 - 180 + 95 = 640$.

TSIA2-M8-014 — Correct: A — Correct Answer: $3x^2 - 13x - 10$ — Explanation: Multiply the binomials: $3x^2 - 15x + 2x - 10 = 3x^2 - 13x - 10$.

TSIA2-M8-015 — Correct: B — Correct Answer: $(-2, 1)$ — Explanation: The midpoint is $((-6 + 2)/2, (4 + -2)/2) = (-2, 1)$.

TSIA2-M8-016 — Correct: A — Correct Answer: Mean — Explanation: Adding 40, which is much larger than the existing values, pulls the mean upward the most. The median changes only slightly, and there is no mode.

TSIA2-M8-017 — Correct: B — Correct Answer: 10 meters — Explanation: The side length is $\sqrt{110}$. Since $10^2 = 100$ and $11^2 = 121$, $\sqrt{110}$ is between 10 and 11 and is closer to 10.

TSIA2-M8-018 — Correct: C — Correct Answer: 5 — Explanation: From $x + 2y = 14$, $x = 14 - 2y$. Substitute into $3x - y = 7$: $3(14 - 2y) - y = 7$, so $42 - 7y = 7$. Then $y = 5$.

TSIA2-M8-019 — Correct: B — Correct Answer: 6 centimeters — Explanation: For a rectangular prism, volume = length $\times$ width $\times$ height. Since $360 = 12 \times 5 \times h$, $360 = 60h$, so $h = 6$ centimeters.

TSIA2-M8-020 — Correct: A — Correct Answer: $\frac{1}{16}$ — Explanation: There are 12 batteries total, and 3 are small. With replacement, the probability of selecting a small battery twice is $\frac{3}{12} \times \frac{3}{12} = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$.